A Size-Independent Discrepancy Bound for Sparse Coverage Functions

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A coverage function counts the features reached by a set of items. Consider any family of such functions in which each item belongs to at most t feature occurrences across the entire family. We prove that, for every integer $k \geq 2$, the items can be partitioned into k color classes so that every function differs by at most $4kt(kt-1)^2$ between any two classes. The bound is independent of the number of items, functions, features, and the sizes of the features. It answers affirmatively the qualitative question of whether sparse coverage discrepancy admits a polynomial bound in t, k alone, motivated by the size-dependent bound of Avila, Rohwedder and Wennmann (ESA 2026). Its dependence on t, k is coarser than theirs.

The proof first uses Hall's theorem to assign disjoint k-item witnesses to all features of size at least kt. Restricting a witness to cyclic color shifts makes every such feature exactly balanced under every allowed choice. A graph on the resulting blocks has at most $(kt-1)^2$ independent classes of small-feature interactions. Within one class, coverage expectations are affine in the block-state probabilities, with maximum column sum at most kt. Applying the known multiple-choice Beck–Fiala rounding theorem class by class proves the bound. The construction is deterministic and polynomial time for explicitly listed incidences, including binary-encoded k and t.

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1 The result and its context

Coverage valuations measure diversity: reaching the same feature twice does not increase its contribution. This saturation prevents a direct application of linear discrepancy bounds. In the sparse setting, however, each item participates in only a bounded number of feature occurrences. We show how to exploit that sparsity without any dependence on the size of the instance.

Here is the exact model. Let n, m, t be positive integers. For each $i \in [m] = \{1, \dots, m\}$, let U_i be a finite feature-index set. A feature occurrence is a tagged pair $e = (i, u)$, with $u \in U_i$, having a nonempty incidence set $S_e \subseteq [n]$. Write $\mathcal{E} = \{(i, u) : i \in [m],\ u \in U_i\}$, and suppose

$$d(j) := \sum_{e \in \mathcal{E}} \mathbf{1}[j \in S_e] \le t \qquad (j \in [n]). \tag{1}$$

The corresponding coverage functions are

$$f_i(T) = \sum_{u \in U_i} \mathbf{1}[S_{(i,u)} \cap T \neq \varnothing] \qquad (T \subseteq [n]). \tag{2}$$

Equal incidence sets at different tags remain distinct occurrences in both sums. Empty features can be removed. The indices U_i may be empty. For a coloring $\chi : [n] \to \mathbb{Z}_k = \{0, \dots, k-1\}$, put $T_c = \chi^{-1}(c)$ and define

$$\operatorname{disc}(\chi) = \max_{i \in [m]} \max_{c, c' \in \mathbb{Z}_k} |f_i(T_c) - f_i(T_{c'})|.$$

Empty color classes are allowed.

Theorem 1.1. *For every family satisfying* (1) *and every integer $k \ge 2$, there is a coloring $\chi : [n] \to \mathbb{Z}_k$ such that*

$$\operatorname{disc}(\chi) \le 4kt(kt-1)^2. \tag{3}$$

For an input that explicitly lists item slots, function and feature tags, and all incidences, such a coloring can be computed deterministically in time polynomial in the input length, with k, t encoded in binary.

In particular, $P(t, k) = 4kt(kt-1)^2$ is one fixed polynomial valid for all n, m, t, k in this model. No bound on feature sizes is assumed.

Closest prior work and the change in parameters. Avila, Rohwedder and Wennmann [ARW26], Theorem 3, prove a randomized polynomial-time bound $O(\sqrt{t^3 k}\log(ntk))$ for sparse coverage functions. Their Section 1.3 discusses avoiding the logarithmic size factor. Our theorem gives a positive answer to the qualitative polynomial-existence question suggested there: every fixed t, k has a bound uniform over all instance sizes. The price is the coarser $O(t^3k^3)$ dependence. We do not establish $O(\sqrt{t^3k})$, nor do we claim a pointwise improvement over their bound for all parameter ranges.

The sparsity convention is essential. We use the covering-universe description in [ARW26, Section 1.1]: each item covers at most t feature elements across all functions. Their dual-set notation in Definition 4 uses a union of collections, which could be read as deduplicating equal incidence subsets. Our explicit tags avoid that ambiguity. The theorem concerns occurrence-counted sparsity as in (1); it does not assert a bound under literal deduplication of repeated subsets.

The technical idea. The proof has two requirements that must hold in the same choice space. Large features must keep all colors throughout rounding, while small features must admit a controlled linearization. Freezing arbitrary colors on representatives can bias small features. Rounding witness items separately can destroy their rainbow property. Instead, we keep each witness as one categorical variable whose states are cyclic shifts. Every state protects its large feature, and the uniform distribution on states treats colors symmetrically. Within a suitable interaction class, a small feature meets at most one moving block, although it may contain several items of that block. Its expectation is still affine in the categorical variable. The crucial column estimate counts the incidences of those items, giving mass at most kt across *all* functions and colors.

Independent interaction classes and affine restricted coverage expectations already appear in [ARW26, Section 5, Lemma 23]. The multiple-choice rounding theorem used here is also known, due to Doerr and Srivastav [DS03]; its exact rational formulation is stated by Glaßer, Reitwießner and Witek [GRW11, Theorem 2]. Hall matching, greedy graph coloring, and that rounding theorem are prior tools. The structural step here is the choice of disjoint rainbow witnesses together with categorical coverage extensions that preserve both requirements and need only a size-independent number of stages.

Broader scope. Hollender, Manurangsi, Meka and Suksompong [HMMS25] and Dupré la Tour and Fujii [DF25] study discrepancy beyond additive functions. The general influence-sparse Lipschitz question in [HMMS25, Definition 13 and Question 14] has a broader function class and a different sparsity measure. The present theorem does not resolve it. Our hypotheses also exclude arbitrary real feature weights and a bound only on the number of influenced functions.

2 Multiple-choice rounding

We give the rounding argument in full to specify exactly the prior result on which the coverage reduction relies. For a finite matrix A, write $\|A\|_1 = \max_a \sum_b |A_{b,a}|$ for its maximum absolute column sum.

Lemma 2.1 (Multiple-choice Beck–Fiala rounding). *Partition a finite set of coordinates into nonempty groups. Suppose $p \geq 0$ sums to one in every group, and A is a real matrix with these coordinates as columns and $\|A\|_1 \leq D$. There is a vector y with exactly one entry equal to one in each group and all other entries zero such that*

$$\|A(y-p)\|_\infty \leq 2D. \tag{4}$$

For rational data and equally sized groups, a suitable y is computable in polynomial time.

The rational equal-group statement and its algorithmic claim are [GRW11, Theorem 2], attributed there to [DS03]. The following floating-coordinate proof establishes the existence statement, also for unequal group sizes; that proof is not claimed as a contribution.

Proof. Start with $x = p$. Fix permanently every coordinate already equal to zero or one. Initially retain every row equation $A_b x = A_b p$. At any stage, let W be the strictly fractional coordinates and put $V = |W|$. If $V = 0$, the process is finished. A group meeting W has at least two coordinates in W, since a nonnegative sum-one group cannot have exactly one fractional coordinate and all others integral. Thus the number G of such groups satisfies $G \leq V/2$.

Drop each retained row b whose residual mass $\sum_{a \in W} |A_{b,a}|$ is at most $2D$. If $D > 0$, the number R of rows still retained satisfies

$$2DR < \sum_{\text{retained } b} \sum_{a \in W} |A_{b,a}| \leq DV,$$

and hence $R < V/2$. If $D = 0$, all matrix entries vanish and all rows have been dropped, so again $R + G < V$. There is therefore a nonzero direction supported on W preserving the retained row equations and the G group-sum equations. Move in that direction until the first coordinate reaches zero or one, and fix all new boundary coordinates. A positive finite move is possible because every coordinate of W starts strictly inside $(0, 1)$. Repeat. At least one additional coordinate is fixed at each move, so there are finitely many moves and the result is an integral group choice y.

A retained row is exact until it is dropped. If row b is dropped at the point x^* with fractional coordinate set W^*, only coordinates in W^* can move later. Consequently

$$|A_b(y-p)| = \left| \sum_{a \in W^*} A_{b,a}(y_a - x^*_a) \right| \leq \sum_{a \in W^*} |A_{b,a}| \leq 2D.$$

A row never dropped remains exact. The error bound charges a row only at its single deletion, rather than at every move, proving (4). The rational running-time claim can be obtained from the cited theorem; Section 6 also describes a rational LP implementation. □

3 Protecting large features and separating small ones

Set $q = kt \geq 2$. Call an occurrence *large* if $|S_e| \geq q$, and *small* otherwise.

Lemma 3.1. *There are pairwise disjoint sets $B_e \subseteq S_e$, each of size k, one for every large occurrence e.*

Proof. For any subfamily Q of large tagged occurrences, count incidences using (1):

$$kt|Q| \leq \sum_{e \in Q} |S_e| \leq t \left| \bigcup_{e \in Q} S_e \right|.$$

Thus $|\bigcup_{e \in Q} S_e| \geq k|Q|$. Form a bipartite graph with k demand copies of every large occurrence on the left and the items on the right; each copy of e is adjacent to S_e. For any subset X of copies, let Q be the occurrences represented in X. Then $|N(X)| = |\bigcup_{e \in Q} S_e| \geq k|Q| \geq |X|$. Hall's theorem gives a matching saturating all copies. Their matched items are the desired witness sets. The argument counts tags even when incidence sets coincide. □

Add one singleton block for each unmatched item. The witnesses and these singletons form a partition $\mathcal{B}$ of $[n]$, with block sizes k or one. Assign distinct base colors $a(j) \in \mathbb{Z}_k$ to the k items of each witness, and base color zero to each singleton. Block B has k allowed states: in state $r \in \mathbb{Z}_k$, its items receive colors

$$\chi(j) = a(j) + r \pmod{k} \qquad (j \in B). \tag{5}$$

In every state each witness contains every color. Every large occurrence therefore contributes exactly one to each color's coverage, for every choice of states of all blocks. It will remain a color-independent constant throughout the proof.

Define a simple graph on $\mathcal{B}$, joining distinct blocks whenever a small occurrence meets both.

Lemma 3.2. *The block graph has a proper coloring with $H \leq (q-1)^2$ classes $C_1, \ldots, C_H$. Each small occurrence meets at most one block in any one class.*

Proof. A block B meets at most $|B|t \leq q$ tagged occurrences. A small occurrence has at most $q-1$ items, so if it meets B it meets at most $q-2$ other blocks. Hence the degree of B is at most $q(q-2)$, where counting neighbors more than once only increases the estimate. Greedy coloring uses at most $q(q-2)+1 = (q-1)^2$ classes. Blocks in one class cannot both meet a small occurrence, by definition of adjacency. When $q = 2$, every small occurrence is a singleton and the graph is edgeless, as the formula requires. □

4 Affine coverage on a class of blocks

Give every block B a probability vector $p_B = (p_{B,r})_{r \in \mathbb{Z}_k}$ on its allowed states. Choose states independently *between blocks*, retaining all dependencies between items of the same block. For the resulting product law define

$$\Phi_{i,c}(p) = \mathbb{E}[f_i(T_c)].$$

Initially let every $p_{B,r} = 1/k$. Adding the same shift to every block rotates all colors and preserves this product law. Thus there is a common value μ_i with

$$\Phi_{i,c}(p^{(0)}) = \mu_i \qquad (c \in \mathbb{Z}_k). \tag{6}$$

This symmetry does not require independent colors within a block.

Fix a class C and hold all probabilities outside C fixed. For a small occurrence e meeting $B \in C$, define

$$z_{e,c}(r) = \mathbf{1}[\text{some } j \in S_e \cap B \text{ has } a(j) + r = c \pmod{k}],$$
$$v_{e,c} = \Pr[\text{no block outside } C \text{ gives color } c \text{ to } S_e].$$

The occurrence meets at most one block of C, by Lemma 3.2. Independence of that block and the outside blocks therefore gives the exact hit probability

$$\Pr[S_e \cap T_c \neq \varnothing] = 1 - v_{e,c} + v_{e,c} \sum_{r \in \mathbb{Z}_k} p_{B,r} z_{e,c}(r). \tag{7}$$

If e meets no class block, its contribution is constant. Summing (7) over small occurrences and adding the constant large-feature contributions gives an affine map

$$(\Phi_{i,c}(p))_{i,c} = b + Ap_C, \qquad A_{(i,c),(B,r)} = \sum_{\substack{e=(i,u)\ \text{small} \\ S_e \cap B \neq \emptyset}} v_{e,c} z_{e,c}(r). \tag{8}$$

The vector b and matrix A depend on the fixed outside laws.

Lemma 4.1. *The matrix in* (8) *has nonnegative entries and* $\|A\|_1 \le q$.

Proof. For fixed B, r, all items of B receive distinct colors. Consequently $\sum_{c \in \mathbb{Z}_k} z_{e,c}(r) = |S_e \cap B|$, including when e contains several items of B. Since $0 \le v_{e,c} \le 1$,

$$\begin{aligned} \sum_{i,c} A_{(i,c),(B,r)} &\le \sum_{\substack{e\ \text{small} \\ S_e \cap B \neq \emptyset}} |S_e \cap B| \\ &= \sum_{j \in B} d_{\text{small}}(j) \le |B| t \le q, \end{aligned} \tag{9}$$

where $d_{\text{small}}(j)$ counts small tagged occurrences containing j. All terms in (8) are nonnegative. This proves the claimed absolute column bound. In particular, multiplicities are included and no extra factor equal to the number of colors or functions is introduced. □

5 Rounding the classes

Proof of the existence assertion in Theorem 1.1. Starting from $p^{(0)}$, process the classes in order. In class C_h, hold all outside laws fixed and apply Lemma 2.1 to (8), with one k-coordinate group per block and $D = q$. Make the states in this class deterministic using the integral group choices returned by the lemma. If $p^{(h)}$ is the full collection of block laws after this step, then for every i, c,

$$|\Phi_{i,c}(p^{(h)}) - \Phi_{i,c}(p^{(h-1)})| \le 2q. \tag{10}$$

At each step the matrix is recomputed using the actual outside laws. Earlier classes are point masses, and later classes are still uniform. Both are legitimate marginals of the same product-law definition of Φ. Thus (10) compares consecutive evaluations of one functional; it does not require the matrices of different classes to coincide.

After H stages every block law is a point mass. The final $\Phi_{i,c}$ is the actual coverage $f_i(T_c)$ of the selected coloring. Telescoping (10) and using (6) gives $|f_i(T_c) - \mu_i| \le 2qH$. Subtracting the same μ_i for two colors and using Lemma 3.2 yields

$$|f_i(T_c) - f_i(T_{c'})| \le 4qH \le 4kt(kt-1)^2.$$

This holds simultaneously for all functions and pairs of colors. □

6 Deterministic construction and encoding length

We justify the time claim in the explicit input model of Theorem 1.1. In particular, a large binary value of k must not force an exponentially expanded state space.

If $k \ge n$, assign distinct colors to the items. A nonempty color class is a singleton, whose value for any f_i is at most $d(j) \le t$; empty classes have value zero. The discrepancy is therefore at most t, which implies (3). The output labels have polynomial length. Hence assume $k < n$. Replace the supplied t, for implementation purposes, by $d = \max(1, \max_j d(j)) \le t$. The function $x \mapsto 4kx(kx-1)^2$ is increasing for $x \ge 1$ and $k \ge 2$, so the resulting guarantee implies (3). Both k and d are now polynomially bounded by the explicit input length. Features and functions with no incidence can be discarded.

Use the threshold kd in Lemma 3.1. The matching graph has at most n demand copies: the proof's counting inequality, applied to all large occurrences, gives this directly. Its edge list is polynomial in the incidence input because $k < n$. Finite bipartite matching, block graph construction, and greedy coloring are

polynomial time. The actual number of nonempty classes is at most the number of blocks, hence at most n, regardless of the numerical upper bound $(kd-1)^2$.

It remains to control the rational matrices used for rounding. At the start of a class, every outside block is either uniform or deterministic. For a uniform outside block B, the probability of missing a fixed color on $S_e \cap B$ equals

$$\frac{k-|S_e \cap B|}{k},$$

because its base colors are distinct. A deterministic outside block has miss probability zero or one. Therefore $v_{e,c}$ is a product of at most n rational factors with denominator k. If $M = |\mathcal{E}|$, every coefficient of (8) has a representation with denominator dividing k^n and bit length $O(n \log k + \log(M+1))$. The initial target for the current uniform class has denominator dividing k^{n+1}. There are at most mk rows and nk columns. The rational matrix and target thus have polynomial encoding length and can be computed exactly. The polynomial-time equal-group version of Lemma 2.1 applies.

For completeness, one may implement the displayed rounding proof without assuming a bit bound for arbitrary repeated nullspace moves. After dropping light rows, consider the bounded rational polytope defined by the remaining row equations, group sums, bounds $0 \le x_a \le 1$, and all previously fixed coordinates. It is nonempty. Since $R+G < V$, a point strictly inside the bounds of all V remaining coordinates cannot be a vertex: a nonzero homogeneous direction would permit movement in both directions. Thus any vertex fixes at least one further coordinate. Obtain a vertex by sequentially optimizing coordinates and restricting to the corresponding optimal faces. Polynomial-time rational linear programming suffices. Each retained row has its original target $A_b p$ for this class, adjusted only by original coefficients of coordinates fixed at zero or one. The constraint coefficients and right sides therefore keep polynomial bit length, and determinant bounds give the same property for vertex coordinates. There are at most nk coordinate fixings per class. This yields a deterministic polynomial-time implementation and completes Theorem 1.1.

7 Limits and verification

The result gives existence and an explicit-input algorithm for all parameters in (1)–(2). It does not optimize the t, k exponents. A single item covering t distinct features has discrepancy t for every $k \ge 2$, the lower-bound example also discussed in [ARW26, Section 1.3]; a substantial parameter gap remains. The categorical construction does not by itself address arbitrary weighted coverage or arbitrary influence-sparse Lipschitz functions.

The proof uses no computational search. Supplementary exact-rational diagnostics tested the categorical identities, witness conditions, and grouped rounding on finite instances. A separate model context also reconstructed the argument and checked finite instances using a separately written exact-arithmetic program. These checks do not establish the infinite theorem, replace the proof, certify priority, or constitute external peer review. The comparison with the cited primary results is limited to the literature inspected.

AI methodology. OpenAI Codex provided substantial assistance with the proof development, counterexample checks, exact-arithmetic code, literature comparison, and the writing of this manuscript; this assistance materially affected all sections. A separate model context assisted with proof and source checks. No claim of human verification or formal proof certification is made.

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